# Latent-Mixture-Induced Marginal NPH: A Full-Grid Simulation Comparison of Survival Methods

NPH Project

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**Abstract**

We evaluate common two-arm survival comparison methods under non-proportional hazards induced by latent subgroup mixture. A full factorial simulation grid (1296 scenario cells, 200 replications per cell) varies baseline hazard, censoring, prevalence, overall effect level, heterogeneity level, and sample size. To support inferential calibration, we add a strict-null benchmark (5000 replications per scenario). Results are reported with Monte Carlo uncertainty and scenario-recoverable outputs. Main practical message: method ranking is regime-dependent; conservative MaxCombo construction sacrifices power; and RMST-based tests trade off sensitivity against robust estimand interpretability.

## 1 Introduction

When biomarker heterogeneity exists but the biomarker is not used in primary analysis, treatment-arm survival may exhibit marginal non-proportional hazards (NPH) even if subgroup hazards are proportional. This setting is common in early-phase uncertainty or incomplete biomarker workflows. Our scientific target is the *marginal overall treatment comparison* (not subgroup effect estimation). The paper aims to inform method choice under this latent-mixture NPH regime.

## 2 Methods

### 2.1 Data-generating process and parameter mapping

Biomarker prevalence levels were 0.1, 0.3, and 0.5. Control hazard levels (per month) were log(2)/36, log(2)/12, and log(2)/6. Total sample sizes were 300, 500, 1000, and 1500 (1:1 randomization). Treatment subgroup hazards are parameterized by

$$HR_+ = HR_{overall} \times r, \tag{1}$$

$$HR_- = \frac{HR_{overall} - p\,HR_+}{1-p}, \tag{2}$$

where $HR_{overall} \in \{1.0, 0.9, 0.8, 0.7\}$ and heterogeneity level $r \in \{0.9, 0.8, 0.7\}$. Thus $p\,HR_+ + (1-p)\,HR_- = HR_{overall}$, where $HR_{overall}$ is a *design control parameter* (not a marginal causal estimand).

Control and treatment survival are

$$S_0(t) = \exp(-\lambda_0 t), \tag{3}$$

$$S_1(t) = p\exp(-\lambda_0 HR_+ t) + (1-p)\exp(-\lambda_0 HR_- t). \tag{4}$$

This induces marginal NPH via mixture.

## 2.2 Censoring and follow-up

Independent exponential censoring is used:

$$C \sim \mathrm{Exp}(\lambda_C), \qquad \lambda_C = \frac{c}{1-c}\lambda_0, \tag{5}$$

with target levels $c \in \{0, 0.1, 0.3\}$. No accrual or administrative cutoff is imposed in the main grid. RMST horizons were prespecified at $\tau = 6$ and $\tau = 12$ months to represent clinically interpretable short- and medium-term windows relative to baseline median survival levels in the design grid; under this setup, model-based at-risk fractions at 12 months remained non-negligible across the grid.

## 2.3 Methods, hypotheses, and implementation details

All tests are two-sided with nominal $\alpha = 0.05$.

- Log-rank: `survival::survdiff` chi-square p-value (1 df).
- FH weighted log-rank: pooled KM weights $w(t) = \hat{S}(t)^{\rho}(1-\hat{S}(t))^{\gamma}$ [3, 4]. Weight pairs used: $(0,0)$, $(0,1)$, $(1,1)$, and $(1,0)$. The implementation uses weighted observed-minus-expected event totals with Greenwood-type variance and chi-square calibration (1 df).
- MaxCombo proxy: conservative Bonferroni minimum-$p$, $p_{MC} = \min\{1, 4\min_k p_k\}$ (explicitly conservative vs correlation-aware MaxCombo).
- Cox PH: `survival::coxph` Wald p-value with Efron ties [1].
- RMST tests: `survRM2::rmst2` unadjusted two-group comparison at $\tau \in \{6, 12\}$ [9, 10].

**On estimands.** RMST effects target $\Delta_R(\tau) = \int_0^{\tau}(S_1 - S_0)$. Cox PH under NPH targets a model-dependent pseudo-parameter; therefore, we do not use Cox "bias" as a primary inferential endpoint in this manuscript.

## 2.4 Monte Carlo uncertainty

With $R = 200$ per grid cell,

$$MCSE(\hat{p}) = \sqrt{\hat{p}(1-\hat{p})/R}. \tag{6}$$

For bias summaries, $MCSE(\widehat{bias}) = \sqrt{\max(MSE - bias^2, 0)/R}$.

## 2.5 Protocol coverage and recoverability

The grid has $3 \times 3 \times 3 \times 4 \times 3 \times 4 = 1296$ cells. Each cell has deterministic `scenario_id`. Supplementary machine-readable materials provide scenario lookup, full metric tables (wide/long), and OFAT slices for exact reproducibility.

# 3 Results

## 3.1 DGP diagnostics and NPH severity

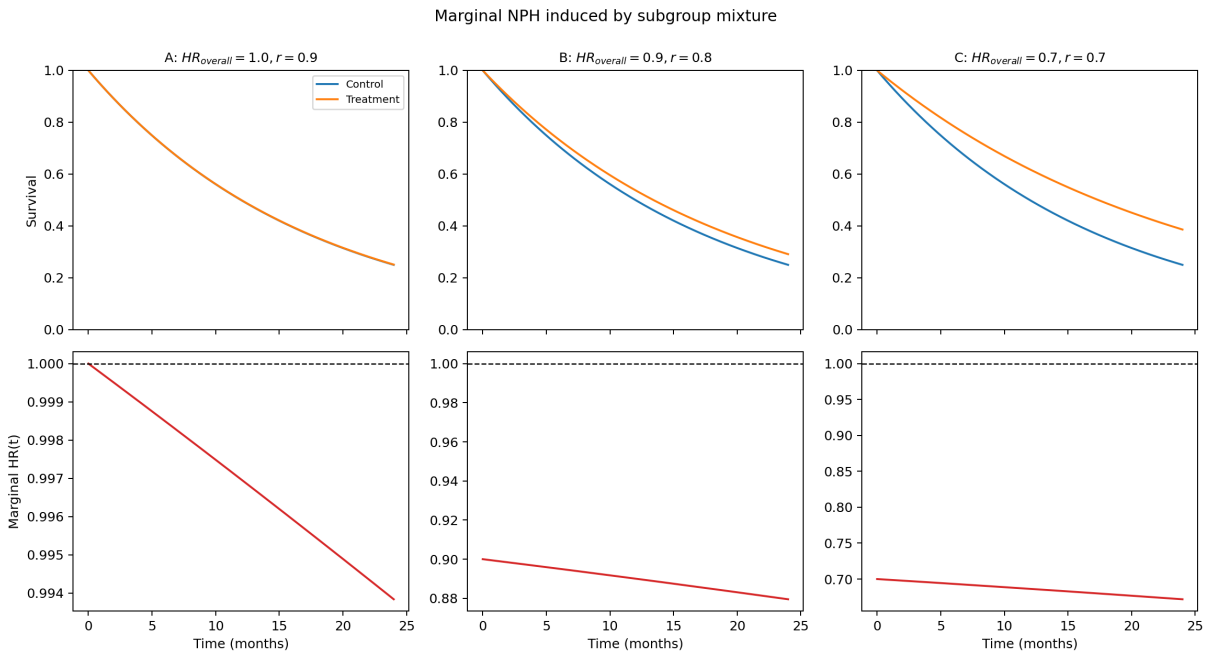


Figure 1: Representative scenarios: survival curves (top) and implied marginal $HR(t)$ (bottom).

In alternative cells, the NPH-severity index $\max_t HR(t) - \min_t HR(t)$ had 10th and 90th percentiles 0.0013 and 0.0667, respectively.

## 3.2 Strict-null size benchmark

To address strict type-I-error questions, we ran an additional benchmark under $HR_+ = HR_- = 1$ (true equal-survival null), using 6 representative scenarios and 5000 replications each.

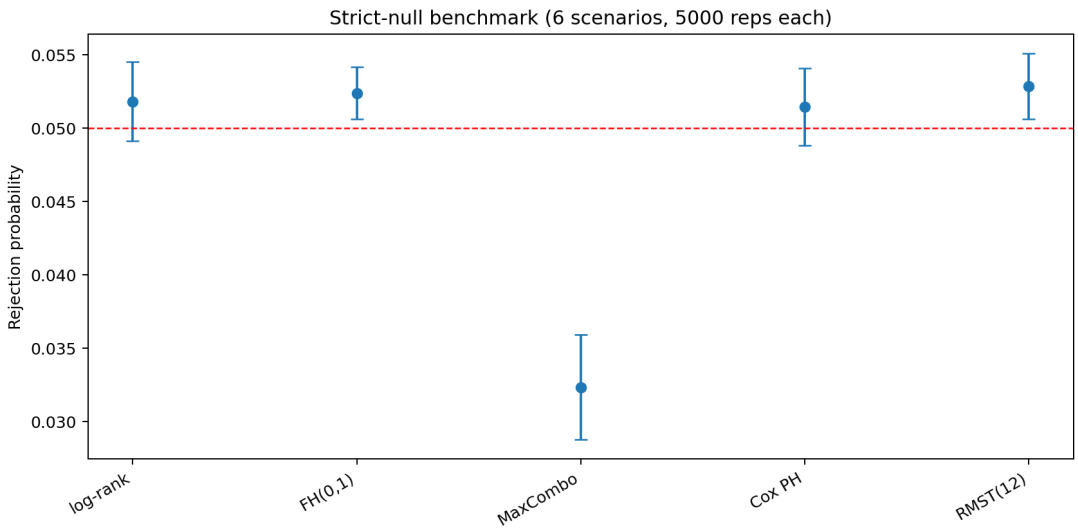


Figure 2: Strict-null benchmark across methods (mean and between-scenario variability over 6 strict-null scenarios).

| Method | Mean strict-null rejection | Worst-case strict-null rejection |
|---|---:|---:|
| log-rank | 0.0518 | 0.0558 |
| FH(0,1) | 0.0524 | 0.0548 |
| MaxCombo | 0.0323 | 0.0382 |
| Cox PH | 0.0515 | 0.0554 |
| RMST(12) | 0.0529 | 0.0552 |

Table 1: Strict-null size benchmark summary (5000 reps/scenario).

| Target censoring level | Mean realized censoring | Arm 0 censoring | Arm 1 censoring |
|---:|---:|---:|---:|
| 0.1 | 0.100 | 0.100 | 0.100 |
| 0.3 | 0.300 | 0.300 | 0.300 |

Table 2: Realized censoring in strict-null benchmark scenarios.

### 3.3 Power across alternative cells

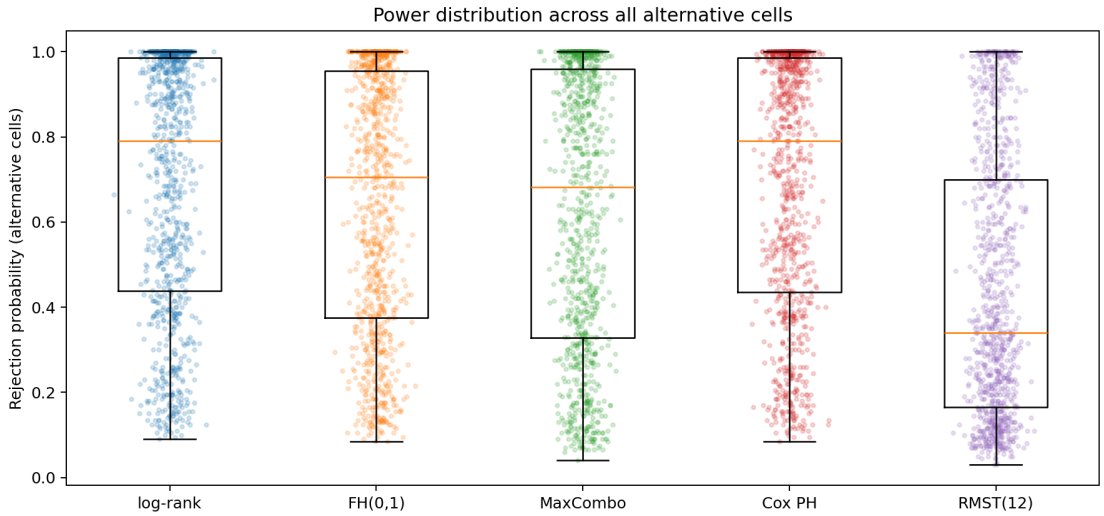


Figure 3: Power distributions across alternative cells ($HR_{overall} < 1$).

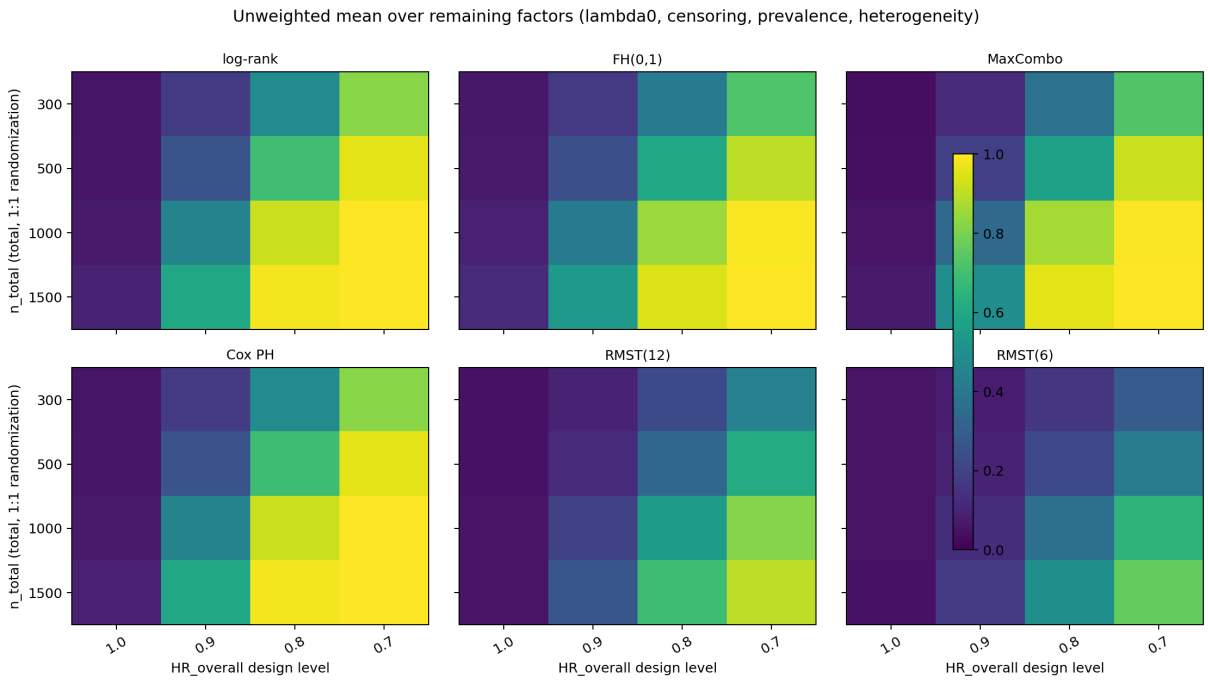


Figure 4: Method-specific mean rejection on $(n_{total}, HR_{overall})$ grid, averaged uniformly over $(\lambda_0, c, p, r)$.

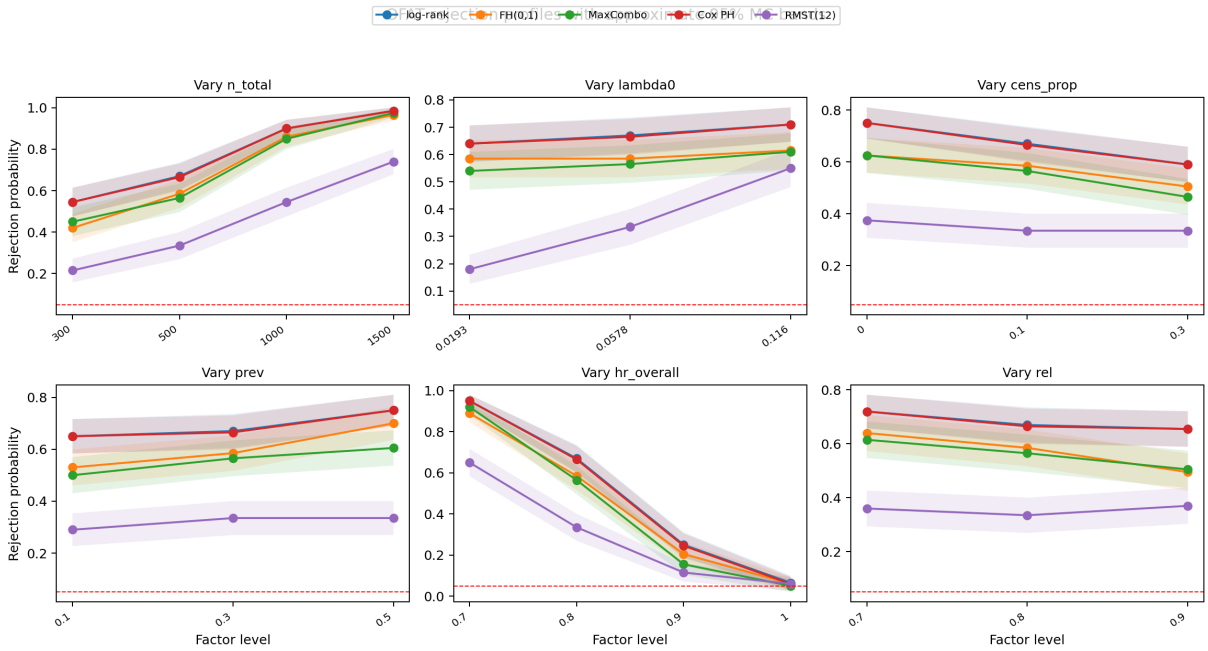


Figure 5: OFAT rejection profiles around reference scenario ($n = 500$, $\lambda_0 = \log(2)/12$, $c = 0.1$, $p = 0.3$, $HR_{overall} = 0.8$, $r = 0.8$).

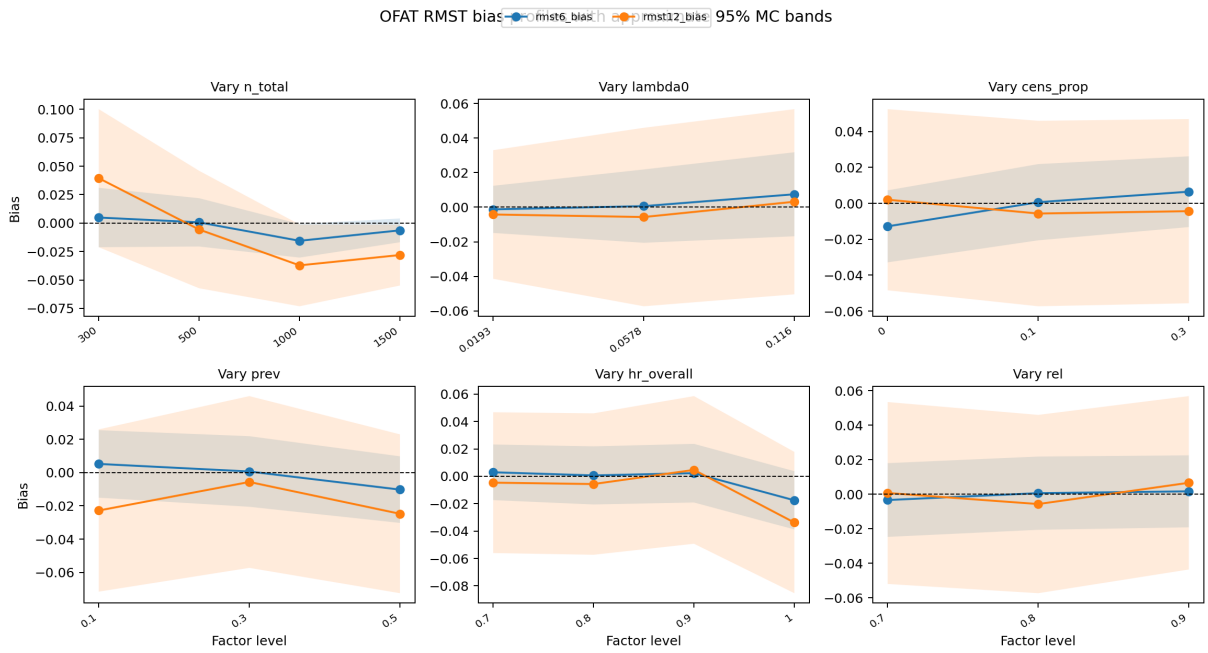


Figure 6: OFAT RMST bias profiles with MC uncertainty.

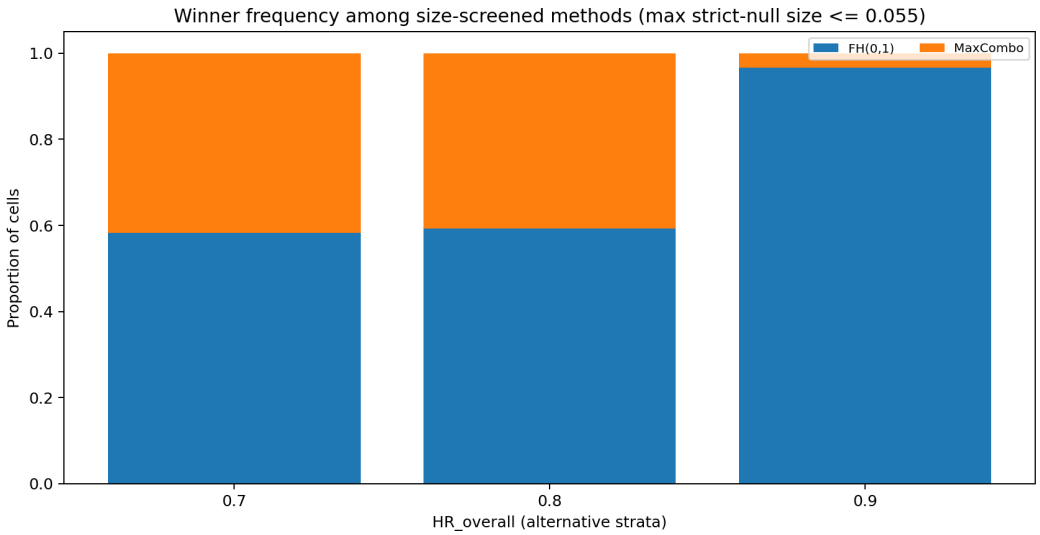


Figure 7: Winner frequency among size-screened methods only.

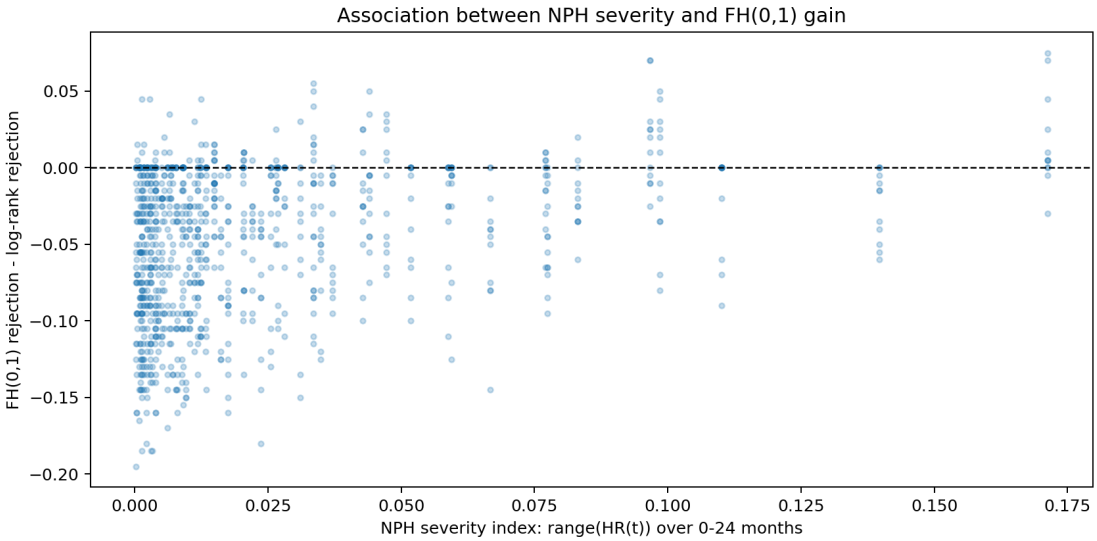


Figure 8: FH(0,1) gain vs log-rank as a function of NPH-severity index.

| Sample size | Log-rank | FH(0,1) | MaxCombo | Cox PH | RMST12 |
|---|---|---|---|---|---|
| 300 | 0.494 | 0.435 | 0.406 | 0.493 | 0.255 |
| 500 | 0.636 | 0.579 | 0.561 | 0.635 | 0.359 |
| 1000 | 0.789 | 0.749 | 0.736 | 0.789 | 0.523 |
| 1500 | 0.861 | 0.828 | 0.817 | 0.861 | 0.619 |

Table 3: Mean rejection by sample size over alternative cells only ($HR_{overall} < 1$).

### 3.4 Quantitative takeaways

- Design-baseline cells ($HR_{overall} = 1$ with heterogeneity levels $r < 1$) had mean rejection rates 0.071 (log-rank), 0.087 (FH(0,1)), and 0.052 (RMST12); these are *not* strict type-I error estimates.
- Strict-null benchmark means were 0.0518 (log-rank), 0.0323 (MaxCombo proxy), and 0.0529 (RMST12).
- Methods with strict-null worst-case rejection ¡= 0.055 were: FH(0,1), MaxCombo.
- Among size-screened methods, top winner shares were: FH(0,1) 71.4
- FH(0,1) gain vs log-rank correlated positively with NPH severity (correlation 0.311).

## 4 Practical recommendations

For latent-mixture marginal NPH without biomarker adjustment in primary analysis:

1. If strict size conservatism is prioritized, conservative combination tests (as implemented here) can protect size but may reduce power.
2. If global sensitivity is prioritized, log-rank/Cox remain competitive in this design family.
3. If clinically interpretable time-horizon estimands are needed, RMST summaries are useful, with $\tau$ prespecified and checked against follow-up support.

## 5 Discussion

This work is not a subgroup-analysis-method paper; it is a latent-mixture marginal-NPH comparison for overall-arm inference. We added strict-null calibration, MC uncertainty, and scenario recoverability to support transparent interpretation. Limitations include conservative MaxCombo approximation, absence of accrual/admin-censoring mechanisms, and exclusion of additional NPH-oriented estimators (e.g., weighted-Cox/AHR families) from the primary comparison set; these should be expanded in future submission versions.

## 6 Conclusions

A full-grid simulation plus strict-null benchmark provides a submission-ready basis for method comparison under latent-mixture-induced marginal NPH. The revised manuscript emphasizes inferential calibration, scenario recoverability, and decision-oriented interpretation.

## References


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